

Summer Class: Machine Learning and Data Science

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Recap: Finding the right basis

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- What if we used a fixed, universal set of basis vectors?
 - Fourier Analysis.
 - The fundamental principle: Any complex signal can be perfectly represented by adding up simple sine and cosine waves of different frequencies and amplitudes.

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- L : The fundamental period of the function.
- n : The harmonic number $(1, 2, 3, \dots)$.
- a_0, a_n, b_n : The Fourier coefficients that determine the weight of each sine and cosine term.

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This allows us to isolate each coefficient one by one.

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- ② Integrate over the period $[0, L]$. Due to orthogonality, every term on the right side becomes zero except for the single cosine term where $n = m$.

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MATLAB Demonstration. 

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- What happens when a Fourier series tries to represent a function with a sharp jump (a discontinuity)?
 - Square wave function has a jump discontinuity at $x = 0$.
 - The series uses an infinite number of sine waves to approximate the sharp corners.
- The series attempts to model the vertical line, but since it's made of smooth sine waves, it struggles. This struggle results in a peculiar and persistent overshoot at the point of the jump.
- No matter how many terms you add to the series, this overshoot never disappears.

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The Fourier exponential series is a more compact and often mathematically convenient way to write a Fourier series.

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This single integral replaces the three separate integrals required for a_0 , a_n and b_n in the trigonometric form.

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$c_n \rightarrow a_n, b_n$:

- $a_0 = 2c_0$
- $a_n = c_n + c_{-n} = 2\text{Re}(c_n)$
- $b_n = i(c_n - c_{-n}) = -2\text{Im}(c_n)$

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- In essence, it answers the question: “What is the recipe of frequencies that makes up this specific signal?”
 - Time Domain: How a signal's amplitude changes over time.
 - Frequency Domain: Which frequencies are present in the signal.

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This shows that even a simple rectangular pulse is actually composed of a wide, continuous range of frequencies, with the most dominant frequencies centered around zero. The “sidelobes” of the sinc function represent the higher frequencies needed to create the sharp edges of the pulse.

Fourier Transform

- Forward Transform (Time to Frequency): Maps a time-domain signal $f(t)$ to its frequency-domain representation $F(\omega)$.

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- t represents time.
- ω represents angular frequency $\omega = 2\pi f$.

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- Convolution Theorem: A complex convolution operation in the time domain becomes a simple multiplication in the frequency domain.

$$\mathcal{F}[f(t) * g(t)] = F(\omega) \cdot G(\omega)$$

Discrete Fourier Transform (DFT)

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In simple terms, the DFT takes a sequence of data points (like an audio sample or sensor readings) and tells you which frequencies are present in that sequence. It converts data from the time domain to the frequency domain.

Unlike the continuous transform, the DFT works with a specific number of discrete data points and outputs the same number of discrete frequency components.

Discrete Fourier Transform (DFT)

- We have a finite sequence of N data points, $x_0, x_1, \dots, x_{N-1}$. The sequence is typically sampled from a continuous signal $f(t)$ at a sampling interval Δt , so $x_n = f(n\Delta t)$.

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- Forward DFT (Time to Frequency):

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i \frac{2\pi kn}{N}} \quad \text{for } k = 0, 1, \dots, N-1$$

- n is the time-domain index.
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- The formula for the elements of the matrix is:

$$W_{kn} = \omega^{kn}$$

where $\omega = e^{-i\frac{2\pi}{N}}$ is a primitive N -th root of unity. The indices k and n range from 0 to $N - 1$.

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Fast Fourier Transform (FFT)

- The Problem: Direct computation of the DFT involves N complex multiplications and $N - 1$ complex additions for each of the N coefficients X_k . This results in a computational complexity of $\mathcal{O}(N^2)$. For a million data points, this is a trillion operations.

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- The FFT is an algorithm that reduces this complexity to $\mathcal{O}(N \log N)$. The most common version is the Cooley-Tukey algorithm, which works by recursively breaking a DFT into smaller DFTs.

Fast Fourier Transform (FFT)

- The Core Mathematical Insight (Danielson-Lanczos Lemma): If N is composite (e.g., $N = 2M$), a DFT of size N can be split into two DFTs of size M . The original sequence $\{x_n\}$ is split into its even-indexed and odd-indexed elements:
 - $x'_m = x_{2m}$ (even part)
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$$X_k = X'_k + e^{-i\frac{2\pi k}{N}} X''_k \quad \text{for } k = 0, \dots, M-1$$

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This process is applied recursively. The term $e^{-i\frac{2\pi k}{N}}$ is called a twiddle factor. By intelligently reusing the results of the smaller DFTs, the FFT drastically cuts down on redundant calculations.

Machine Learning Based on Fourier Analysis

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Fourier analysis isn't a machine learning algorithm itself, but it's one of the most powerful tools for feature engineering in signal processing.

- Reveals Hidden Patterns: It makes frequency-based characteristics explicit and measurable.
- Reduces Dimensionality: It converts thousands of raw data points into a handful of potent features, making the ML model faster and more efficient.

Machine Learning Based on Fourier Analysis

Raw time-series data, like audio or sensor readings, often hides the very patterns we need for machine learning. Two signals can look very similar in the time domain but be fundamentally different in their underlying frequency content.

- Time Domain: Shows a signal's amplitude over time. It can be noisy and complex, making classification difficult.
- Frequency Domain: Shows which frequencies are present in the signal and at what strength. This often reveals a clear “signature” for each class.

Machine Learning Based on Fourier Analysis

We use the Fast Fourier Transform (FFT) to convert a signal from the time domain into the frequency domain. From the resulting spectrum, we extract specific, measurable features to feed into our machine learning model.

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- Peak Amplitude & Frequency: The strength and location of the most dominant frequency.
- Band Power: The total energy within a specific frequency range.
- Spectral Centroid: The “center of mass” of the spectrum. A high centroid means more high-frequency energy.
- Frequency Ratios: The ratio of power between different bands.

Machine Learning Based on Fourier Analysis

MATLAB Demonstration: Structural Health Monitoring.

Summary of What You've Learned Today

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- Fourier Transform.
- Discrete Fourier Transform.
- Fast Fourier Transform: Extract Features from Time Series.

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See you on Wednesday (7/9) for Lecture 5 on
Linear Regression.